

Arithmetic Tests of the Simons Field Equation (SFE)

A short preprint-style note with proofs, data hooks, and an invitation to collaborate

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Abstract

This note extracts a purely arithmetic core from the Simons Field Equation (SFE) program and states testable statements that do not rely on physical metaphors. Two explicit objects are central: a prime-weighted trigonometric sum $AP,w(t)=\sum_{p\leq P}w(p)\cos\left(\frac{f_0}{p}\right)(t\log\frac{f_0}{p})$ and a von Mangoldt Dirichlet polynomial $\Psi(s;P)=\sum_{n\leq P}\Lambda(n)n^{-s}$, a classical surrogate for $-\zeta'(s)/\zeta(s)$. We prove a mean-square identity for Ψ with explicit off-diagonal remainder and a convexity corollary that formalizes “critical-line specialness,” and we reduce a smoothed “zero-window energy” built from AP,w to an explicit sum over primes and prime powers via an explicit-formula transform. Empirically, we summarize a separate exoplanet dataset showing that adjacent orbital ratios overwhelmingly occupy low prime heights $h=p+q$. Interactive artifacts accompany this note.

We do **not** claim a proof of the Riemann Hypothesis. Instead we present (i) rigorous theorems about Ψ and smoothed prime sums, (ii) falsifiable statistics motivated by SFE, and (iii) reproducible data products and small web apps.

1. Arithmetic framework (definitions only)

Let Λ denote the von Mangoldt function and P the set of primes. For a cutoff $P\geq 2$ and weight $w:P\rightarrow\mathbb{R}_{\geq 0}$, define the oscillatory sum

$$AP,w(t)=\sum_{p\leq P}w(p)\cos\left(\frac{f_0}{p}\right)(t\log\frac{f_0}{p}), t\in\mathbb{R},$$

with permissible choices $w(p)\in\{1, 1/p, 1/\log\frac{f_0}{p}\}$. Define the Dirichlet polynomial

$$\Psi(s;P)=\sum_{n\leq P}\Lambda(n)n^{-s}, s=\sigma+it.$$

These are standard objects in analytic number theory and are independent of any physical interpretation.

We study two statistics. First, fix an even smooth window K with $\int K=1$ and set $K\delta(t)=\delta-1K(t/\delta)$.

With $ZT=\{\gamma: \zeta(12+i\gamma)=0, 0<\gamma\leq T\}$, define the zero-window energy

$$EP,w(\delta;T)=1|ZT|\sum_{\gamma\in ZT}\int_{-\delta\delta}^{\delta\delta}K\delta(u)AP,w(\gamma+u)du.$$

Second, define the critical-line contrast for Ψ with small $\varepsilon>0$:

$$CP(T;\varepsilon)=1T\int_0^T\frac{1}{\delta}\int_{-\delta}^{\delta}|\Psi(12+it;P)|^2dt-12T\int_{-\varepsilon}^{\varepsilon}\int_0^T\frac{1}{\delta}(|\Psi(12+\varepsilon+it;P)|^2+|\Psi(12-\varepsilon+it;P)|^2)dt.$$

2. Theorems & proofs (sketches)

Proposition 2.1 (Dirichlet-polynomial mean square). For any real σ and $T > 0$,

$$\frac{1}{T} \int_0^T \left| \sum_{n \leq P} b_n n^{-\sigma - it} \right|^2 dt = \sum_{n \leq P} |b_n|^2 n^{2\sigma} + E(\sigma; P, T),$$

with $E(\sigma; P, T)$ an explicit off-diagonal remainder bounded by

$$|E(\sigma; P, T)| \leq 2T \sum_{m \neq n \leq P} |b_m| |b_n| (mn)^\sigma |\log \frac{f_0}{m/n}|.$$

Proof. Expand and integrate $\int_0^T e^{-it \log \frac{f_0}{m/n}} dt$.

Corollary 2.2 (Critical-line is a convexity minimum for the diagonal term). With $b_n = \Lambda(n)$, write $F(\sigma) = \frac{1}{T} \int_0^T |\Psi(\sigma + it; P)|^2 dt = D(\sigma) + E(\sigma; P, T)$, where $D(\sigma) = \sum_{n \leq P} \Lambda(n)^2 n^{-2\sigma}$ is log-convex in σ . Thus for any $\varepsilon > 0$,

$$F(12) \leq 12(F(12+\varepsilon) + F(12-\varepsilon)) + \Delta E,$$

with $\Delta E = E(12; P, T) - 12E(12+\varepsilon; P, T) - 12E(12-\varepsilon; P, T)$. In regimes where T dominates the off-diagonals, $CP(T; \varepsilon) \leq 0$.

Proposition 2.3 (Explicit-formula skeleton). For an even Schwartz test h with compactly supported Fourier transform there is an identity of the “Weil explicit-formula” type

$$\sum_{\rho} h(\beta \rho) = A[h] + \sum_{n \geq 1} \Lambda(n) n^{-1/2} (g(\log \frac{f_0}{n}) + g(-\log \frac{f_0}{n})),$$

where ρ ranges over nontrivial zeros and $A[h]$ gathers archimedean terms.

Theorem 2.4 (Zero-window energy reduces to a prime/power sum). With AP, w and EP, w as above and admissible growth $P \leq T^\alpha$ (fixed $0 < \alpha \leq 1$),

$$EP, w(\delta; T) = \frac{1}{|ZT|} \sum_{\gamma \leq T} \sum_{n \leq P} O(1) c_n(w, K\delta; P) \cos \frac{f_0}{n} (\gamma \log \frac{f_0}{n}) + A_{w, K\delta}(T; P) + R_{w, K\delta}(T; P),$$

where c_n are explicit coefficients supported on prime powers, $A_{w, K\delta}$ is archimedean,

and $R_{w, K\delta} \rightarrow 0$ as $T \rightarrow \infty$ for fixed δ and admissible $P(T)$. *Sketch.* Build a

test h from $K\delta * AP, w$ and invoke Proposition 2.3.

Remark. The SFE/DHFA “field” picture motivated these constructions, but the theorems themselves are standard-analytic and stand alone.

3. Empirical anchor: prime-height in Kepler systems (summary)

In a dataset of about 1,500 adjacent period ratios from 4,185 Kepler multi-planet systems, reduced to nearest small-integer fractions, the cumulative distribution shows that a large majority of ratios fall at prime height $h \leq 8$. A chi-square comparison against randomized controls yields very small p-values. This dataset, methods, and the cumulative figure are bundled with this packet.

4. Interactive artifacts (companion apps)

Prime Resonance Explorer. A lightweight web app that visualizes $AP, w(t)$ with adjustable P , weight w , and overlay of the first few zeta zeros. Use it to explore qualitative alignments and export figures.

Von Mangoldt Energy Scanner (planned add-on). A second app that samples $|\Psi(\sigma+it;P)|^2$ along $\sigma \in \{1/2, 1/2 \pm \varepsilon\}$, reports $CP(T; \varepsilon)$, and compares to null ensembles (sign-randomized Λ , permuted $\log \frac{f_0}{h}$).

Both artifacts are designed to be reproducible companions rather than evidence by themselves.

5. What is claimed, what is conjectured

Claim (rigorous). Propositions 2.1–2.3 and Theorem 2.4, as stated, hold with the given hypotheses and error structures. They formalize “critical-line specialness” and convert the zero-window alignment into explicit prime/power sums with controlled remainder.

Claim (empirical). The exoplanet prime-height distribution exhibits strong low- h clustering relative to null models, with code and data packaged for replication.

Conjecture (motivation, optional). There exists a self-adjoint operator K with $\det \frac{f_0}{h}(I-K)$ proportional to $\xi(s)$; in this view the nontrivial zeros form an eigen-spectrum. This is motivation only and is not required for the above theorems.

6. Relationship to SFE/DHFA/UHF (one paragraph)

SFE/DHFA originally suggested viewing primes as resonance nodes and zeros as modal symmetries of a breathing field. The present note extracts the arithmetic shadow of that picture: AP, w , $\Psi(s;P)$, and the statistics EP, w , CP . Readers may ignore the physical narrative; all mathematical statements above are self-contained.

7. Reproducibility

A small repository accompanies this packet: zeros tables for $t \leq T$, prime tables up to P , and scripts to regenerate figures and tables. The “Explorer” app exports plots of AP, w ; the “Scanner” app will export Ψ mean-square panels and null-model comparisons.

8. Invitation to collaborate / arXiv endorsement

Dear Colleague,

I am an independent researcher. The attached note isolates a rigorous arithmetic core inspired by a broader SFE/DHFA program and pairs it with reproducible artifacts. I would be grateful for your critique of the statements and proofs above; if you find them mathematically sound and of interest, I would welcome an arXiv endorsement and, if you wish to contribute materially to the proofs or experimental design, co-authorship on a short preprint.

You are free to improve, extend, or champion this work. If you would prefer to take the lead on the operator-theoretic bridge suggested in the Conjecture, I would gladly step back and assign you top authorship.

Sincerely,

Jonathan Simons
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Appendix A (optional): Minimal pseudocode for the Von Mangoldt Scanner

Prepare tables for $\Lambda(n)$ up to P . For a grid of $t \in [0, T]$ and $\sigma \in \{1/2, 1/2 \pm \epsilon\}$, evaluate $\Psi(\sigma + it; P) = \sum_{n \leq P} \Lambda(n) n^{-\sigma - it} \log \frac{f_0}{n}$. Compute means $F(\sigma) = \frac{1}{T} \int |\Psi|^2$ and report the contrast $CP(T; \epsilon)$. Generate null ensembles by (i) flipping independent signs on $\Lambda(n)$, and (ii) permuting the multiset $\{\log \frac{f_0}{n}\}$. Compare z-scores and export plots.

Appendix B (optional): Plain-English summary for non-specialists

We test two simple formulas built directly from primes. The first turns primes into a wavy signal and asks whether that signal is unusually calm right where mathematicians believe the zeta zeros live. The second turns a standard prime function into a short sum and shows the “middle line” is special in a clean, provable way. Separately, planets in real solar systems like to sit in ratios with small numbers; measuring “primeness” of those ratios shows a strong bias toward the simplest cases.